

QUAD

Qualcomm Unified Agent for Developers

The CUDA Killer from Qualcomm

Executive Pitch | May 2026 | Qualcomm Confidential

The Problem

Developer Friction is Killing Qualcomm Adoption

SDK Sprawl — 5 SDKs (QNN, SNPE, Hexagon, Adreno, AIMET) — separate install, API, docs

Manual Profiling — Launch 3+ profiling tools separately for CPU/GPU/NPU

Platform Fragmentation — Different workflows per chipset (Windows vs Android vs IoT)

Days-Long Setup — TTFI takes days due to environment configuration

The Solution

One Platform. All Qualcomm Silicon.

Layer	Component	CUDA Equivalent
7 — DevX	CLI + IDE + AI Agent (MCP)	— (unique)
6 — Serve	Inference Server + Model Zoo	Triton
5 — Optimizer	Graph Fusion + Quantization	TensorRT
4 — Libraries	QualcommDNN + QualcommBLAS	cuDNN + cuBLAS
3 — Runtime	Device + Tensor + Model + Stream	CUDA Runtime
2 — Compiler	QUAD IR + Portable Binary	nvcc + PTX
1 — Hardware	CPU + GPU + NPU	GPU only

Why Now

The Market Window is Open

45 TOPS NPUs — Match data center accelerators for inference workloads

On-Device AI Explosion — Privacy + latency + offline → Qualcomm owns this space

NVIDIA Has No Edge — But developers still default to CUDA due to tooling gap

"Qualcomm ships 3+ BILLION SoCs/year. CUDA reaches millions. QUAD reaches billions."

What We Built So FAR

Feature-Complete Platform (Mock Mode)

933 Tests | 15 Modules | 30+ Templates | 15,000+ Lines

Runtime	Compiler	Libraries	Optimizer	Profiler
Kernels	Serve	CLI	UDO	PSNPE
Adapters	Platforms	Codegen	Models	Utils

Where QUAD Wins

vs NVIDIA CUDA

Dimension	CUDA	QUAD
Market	Data center / HPC	Edge / Mobile / AI PC / IoT
Compute	GPU only	CPU + GPU + NPU (heterogeneous)
Power	300-700W	1-15W (20x better perf/watt)
Scale	Millions of GPUs	Billions of SoCs worldwide
Differentiator	Raw throughput	Performance-per-watt + AI agent
Developer UX	Manual SDK	Natural language (MCP)

What CUDA Can Never Do

Power-Aware Computing — Every API accepts power budget. Battery life is a feature.

AI Agent (MCP) — "Convert to INT8 and deploy to phone" → production code in seconds

Heterogeneous Orchestration — CPU+GPU+NPU allocation in one call

Edge-Native — Designed for 1-15W with thermal-aware scheduling

Success Metrics

Metric	Target	Timeline
TTFI	< 5 minutes	Q3 2026
Developer NPS	≥ 60	Q4 2026
pip install $\rightarrow$ inference	Works	Q2 2026
StackOverflow questions	> 1,000	Q4 2026
Model Zoo	100+ models	Q3 2026
Enterprise deployments	≥ 10	Q1 2027

The Ask

- 1. SDK Documentation** — Unblocks real adapter in 2-3 days per SDK
- 2. Hardware (3 devices)** — Snapdragon X Elite, Arduino UNO Q, Snapdragon 8 Elite
- 3. Engineering (12-15 people)** — 12 months to CUDA parity; differentiators from Day 1

Phase 1 (now): SDK docs → real adapter | Phase 2: expand team | Phase 3: community launch

Live Demo Available

Mock Mode (today) — ./install.sh → 933 tests pass, all tools functional

Claude Code — Open project → QUAD MCP auto-detected → natural language AI

Full Pipeline — quad quickstart → detect → compile → profile → generate

```
git clone git@github.qualcomm.com:pavanr/QUAD.git && cd QUAD
```

```
./install.sh && source activate.sh && quad quickstart
```

The Opportunity

3 billion Qualcomm SoCs per year.

Zero unified developer platform.

**QUAD makes Qualcomm the default choice
for on-device AI development.**

`pip install qualcomm-ai-toolkit && quad quickstart`

Thank You

**QUAD — The Agent for Developers for
Qualcomm**

[github.qualcomm.com/pavanr/QUAD](https://github.com/Qualcomm/pavanr/QUAD)